CARSEM

ENGINEERING CHANGE NOTICE

E.C.N. NO : 2512018M PAGE 1 OF 163

TITLE : CONTROL OF NON-CONFORMING MATERIAL/PRODUCTS

SPECIFICATION NO : CRQA100H07 | REVISION : CN

[✓] PERMANENT
[ ] NEW SPECIFICATION
[✓] SPECIFICATION REVISION
[ ] REVIEW FOR COMPLIANCE
[ ] OBSOLETE SPECIFICATION

[ ] TEMPORARY ECN
[ ] *PERMANENT CHANGE
[ ] TEMPORARY CHANGE

*PERMANENT CHANGE TO BE ENCLOSED WITH ECN TO SPEC DEPARTMENT THROUGH E-MAIL.

TRAINING NEEDS:
(TO BE FILLED BY ORIGINATOR)

[✓] ORIGINATOR REQUIRE UPDATE TRAINING MANUAL INFORMATION AND CONDUCT TRAINING
[ ] TRAINING NOT REQUIRED
[ ] TRAINING TO BE DONE BY TRAINING DEPT AND REQUIRE TO UPDATE THE TRAINING MANUAL
[ ] JUSTIFICATION IF NO REQUIRE TO UPDATE TRAINING MANUAL CHANGES NOT RELATED TO THE PROCESS

TRAINING DEPT'S ACKNOWLEDGEMENT : **MAHANI / KARIMUDDIN / CHEN XH**

JUSTIFICATION FOR CHANGE :
NOTE : - THOSE MARKED * MUST REQUIRE ECN TO BE RAISED OTHERWISE TECN WILL BE REJECTED.
EXCEPTION IS CONSIDERED PROVIDED EVIDENCE ATTACHED FOR THOSE ITEM MARKS '*' ONLY.

[ ] * QUALIFICATION BUILD
[ ] AUDIT RESPONSE
[ ] NEW MACHINE / PACKAGE / MATERIAL / PROCESS
[ ] NEW SPECIFICATION / PROCEDURE
[ ] PROCESS / QUALITY IMPROVEMENT
[✓] 8D ACTIONS, CARSEM REF # : SRENC2025113526
[ ] *EVALUATION / RISK BUILD
[ ] SPECIFICATION COMPLIANCE
[ ] 3D ACTIONS, CARSEM REF # :___________________
[ ] OTHERS, PLEASE SPECIFY : ____________________

CHANGE TYPE :
[ ] CUSTOMER REQUEST
[✓] CARSEM INITIATED

THE SPECIFICATION IS APPLICABLE TO :
- CARSEM (M) : [✓] M-SITE [✓] S – SITE
- CARSEM CHINA : [✓] SUZHOU SITE [✓] SUXIANG SITE

TIME FRAME (FOR TECN ONLY)- PLEASE SPECIFY :

____________ WEEK / S or ____________ MONTHS / S

TECN FOR MORE THAN 1 MONTH CAN BE ALLOWED FOR EVALUATION PURPOSE ONLY

EXPIRY DATE : [ ][ ][ ][ ] [ ][ ][ ] [ ][ ]

[ ] NEW [ ] EXTENSION, _____TIME

THE CHANGE(S) IS APPLICABLE TO (FOR TECN ONLY):
- CARSEM (M) : [ ] M – SITE [ ] S – SITE
- CARSEM CHINA : [ ] SUZHOU SITE [ ] SUXIANG SITE

** COUNTER-PART APPROVAL [ ] REQUIRED [ ] NOT REQUIRED

** CHANGES REQUIRED PCN [ ] YES [ ] NO
IF YES , PCN NUMBER : _________________________

*NOTE: ** DETERMINED BY ORIGINATOR*

DESCRIPTION OF CHANGE :
FROM : N/A | TO : AS PER HISTORY OF CHANGES (PAGE 163 OF 163)

RELATED/CROSS REFERENCE SPEC TO BE REVISED:
[✓] NO
[ ] YES. SPEC NO.______________________

ORIGINATOR	LAILY AZWATI
INITIATOR	SK LEE
QA ENGINEER	SHANKRI, SUFIAN, KW LEE, CHANTHIRAVANAN, SURESH, KALAI, AZIZAH, XIA CAIHUA
PROCESS ENGINEER	JUDE, MK KAN, ROSLI, SARAVANAN, YY CHIN, CL CHAN, KF LOH, KW YEE, CL WONG, XIA RONGGUI, WANG MINGREN
ENGINEERING MANAGER	ZAMRI, NAVINDRAN, IRWAN, RICHARD LEE, KF WONG, CHRIS WONG, SHUKRI NIZAM, YK YANG, ANDY XIANG
QA MANAGER	RITA ANN, SY LUM, KT HUM, THAVAKUMARAN, YORK XIAO
QMS MANAGER	REETHA
HEAD OF QUALITY	WAYNE LAI

(TO BE FILLED BY ORIGINATOR)
CUSTOMER APPROVAL : [ ] REQUIRED [✓] NOT REQUIRED
COMPANY NAME : ____________________ NAME / SIGNATURE ____________________

APPROVAL FOR SPECIFICATION RELEASE : SK LEE/ LAILY AZWATI   EFFECTIVE DATE : **2026-JAN-29**

1.0 Title
Control of Non-conforming Material/Products

2.0 Purpose:
To establish and provide guideline and general procedure for quarantine/on-hold, recapture and MRB whenever non-conforming material/products are found.

3.0 Scope:
Apply to all assembly and test areas.

4.0 Reference:
4.1 CRQA100G02 - Document/Records Retention Procedure.
4.2 CRQA100M01 - Corrective and Preventive Action Procedure.
4.3 CRQA100N02 - Customer Complaint and Product Return Procedure.
4.4 QRA000002 – Carsem's Rework Policy.
4.5 QRA000024 – Lot Tolerance Defective (LTPD) Sampling Plan.
4.6 QCG000080 –Guideline for the handling of Internal & Customer Concession Permit.
4.7 CRQA100H03 – Incoming Material Quality Control.
4.8 QCG-073 – Product Scrap page Procedure.
4.9 QCG000017 – Guideline for a Shutdown of an Operation due to Quality Issue.
4.10 IQC00WK02 – Standard Operating Procedure for field failure feedback report to IQA.
4.11 CRQA100H02- Handling, Storage, Packing, Preservation and Delivery.
4.12 QCG000112 - Quality abnormality Management System for Critical Defect.
4.13 IATF 16949:2016 – Automotive Quality Management System Standard.
4.14 ISO 9001:2015 – Quality Management Systems Requirements.
4.15 EMS000024 – Hazardous Substances Management System Manual.

5.0 Responsibility:
5.1 Incoming Personnel:
5.1.1 IQA Inspector:
- To assist IQA supervisor to do physical recapture and containment.
- To perform screening as required.

5.1.2 IQA Supervisor:
- To quarantine and hold non-conforming material in system.
- To ensure recapture and containment of relevant lots are done and result feedback to relevant personnel (e.g. Supplier /IQA Engineer/ Process Engineer).

5.1.3 IQA Engineer:
- To verify defect and also determine sampling plan for recaptured material.
- To coordinate recapture activity of non-conforming product/ material.
- To conduct risk assessment
- To raise a MRB for Material disposition (when the problem was detected at IQA) if customer's waive / disposition required.
- To notify and issue Supplier Corrective Action Request (SCAR) to related supplier.

5.2 Line Personnel:

5.2.1 Operator / Technician

- To notify production supervisor or process technician immediately upon detection of any non-conforming products.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16 ,17,18 and 19).

5.2.2 Production Supervisor

- To confirm/notify QA immediately upon confirmation of the non-conforming material.
- To submit the non-conforming material/products to QA to quarantine the parts at QA Quarantine Cabinet.
- To notify production control and process engineering that the non-conforming material was detected and placed on hold by QA.
- To verify and confirm the non-conformance status of the said product/material.
- To assist QA in carrying out recaptures based on Engineering Recapture instruction and Recapture procedure in the specification.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16 ,17,18 and 19).

5.2.3 Production Superintendent

- Coordinate and ensure Recapture of product has been carried out effectively.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19).

5.2.4 Maintenance Superintendent

- To perform machine and product buy-off with process/ product engineer before released machine for production.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15,\ 16 ,17,18 and 19).

5.2.5 Manufacturing/Product Manager

- To approve the disposition made in MRB after reviewing all pertinent support documents.

5.3 QA Personnel

5.3.1 QA Inspector

- Issue Non-Conformance Report (NCR) to production supervisor if any defects detected at QA Gate.
- To stop QA Gate and notify QA Supervisor/Engineer if found critical defects and 3 times rejection (2 times for automotive product in SZ Site) of the same lot.

5.3.2 QA supervisor / Leader

- To quarantine all affected/ recaptured non conforming material at the QA quarantine area and hold electronically in MES system.
- To coordinate with production to carry out the recapture based on Engineering Recapture instruction and Recapture procedure in the specification.
- To ensure all affected sub-lots are physically pulled back by performing verification of actual on-hold lot # and lot quantity (tally check on system hold report versus original on-hold request list) to ensure no miss out.

- In the event of Machine Shut Down Notice was issued. QA supervisor ensure to send the Machine Shut Down Notice for acknowledgement within 24 hours / 1 working day. If the machine shut down notice issue during night shift, weekend and public holiday, the machine can be release back to production after attend by engineering and the machine shut down notice can be approved on the next working days.
- To release material physically and electronically through the lot tracking system upon receiving documented and signed MRB.
- To issue MRB to production if found critical defects and 3 times rejection (2 times for automotive product in SZ Site) of the same lot at QA Gate.
- Escalate critical defect as defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19)

5.3.3 QA Engineer / Executive
- To verify the root cause established by process engineer
- To ensure recapture procedure is followed strictly as defined by Process Engineer's Instructions and Specification.
- To evaluate the risk of defect encountered, determine the disposition of non conforming material/products and corrective & preventive actions in MRB, together with process engineer.
- To verify and trigger process owner in order to ensure the MRB procedure is complied with.
- Escalate critical defect defined in Quality Escalation Matrix.(Refer to Appendix 15, 16 ,17,18 and 19)
- Lot hold under HQ which was raised MRB, once customer provide disposition to scrap the lot and MRB approved, need to change the hold code to HDCRAP and need to notify QA supervisor to Tag and quarantine in the QA cabinet while waiting for the scrap from approval. (Applicable for S-Site Assembly only)

5.3.4 QA Manager
- To decide if customer notification is required or not together with engineering manager.
- To have final decision on the disposition of Non-conforming Material/Products

NOTE: The QA Personnel is authorized to shut down any operation where non-conforming is cited with the approval from the QA manager (Refer to QCG000017)

5.4 Process Engineer/Process Owner
- To inform QA personnel (supervisor/engineer) immediately upon detection of non conforming material
- To stop affected operation/ process/ material when defect is triggered in production line.
- To determine the cause of failure by performing detailed process mapping.
- To identify lots affected and provide list and all related instruction to production / QA for recapture.

  Noted:Need to include all customer code during reterive risk lots, such as

Customer	Customer code
TI	TIM,TID,TIG,TIT

- To perform risk assessment of the affected lot to determine the extent of the effect with QA Engineer

- To raise MRB, convene a meeting to review the actions necessary to dispose the discrepant parts or lots
- To perform machine/ product buy-off with maintenance before restart production.
- Escalate critical defect defined in Quality Escalation Matrix (Refer to Appendix 15, 16 ,17,18 and 19)

5.5 Engineering Manager
- To ensure Risk Assessment is completed by Engineering and recommend appropriate next course of action on product disposition.
- To decide customer notification for product/ material disposition together with QA manager and lead the communication with customer (if the customer is to be notified).
- To approve the disposition made in MRB after reviewing all pertinent support documents related to the disposition.
- To react to quality alert and ensure corrective & preventive action are taken (As necessary).

5.6 Reliability Engineer
- To perform reliability test for product disposition as per guideline given in the Reliability Test Risk Assessment Matrix (Refer to Attachment C-3).

5.7 PC
- To provide assembly and test WIP report to QA and production in order to perform lot On-hold.

6.0 Material/Equipments:
N/A

7.0 Procedure:

7.1 The procedure for quarantine/on-hold, recapture and MRB of non-conforming material/products shall refer to Appendix 1 to Appendix 14:

7.1.1 Appendix 1: General Flow for Control of Non-conforming Material/Products. (Critical Defect)
7.1.2 Appendix 2: Flow for Control of Incoming Raw Material Failure
7.1.3 Appendix 3: Flow for Control of Line Raw Material Failure Feedback
7.1.4 Appendix 4: Flow for Control of QA Gate Failure
7.1.5 Appendix 5: Flow for Control of Reliability Monitoring Failure
7.1.6 Appendix 6: Flow for Control of Customer Complaint & Field Failure
7.1.7 Appendix 7: Flow for Control of Assembly Low Yield with Critical Defect Failure
7.1.8 Appendix 7A: Flow for Control of Assembly Low Yield/Critical Defect Failure (SZ Site only).
7.1.9 Appendix 8: Flow for Control of Metal Finishing Failure
7.1.10 Appendix 9: Flow for Control of Lead (Pb) Contamination At Plating Process Failure
7.1.11 Appendix 10: Flow for Control of Test & Finishing Failure (Assembly Related Defect)
7.1.12 Appendix 11: Flow for Control of Test Low Yield & High Continuity Lot Verification & Failure
7.1.13 Appendix 12: Flow for Control of Test and Reel Failure
7.1.14 Appendix 13: Flow for Control of measuring equipment / instruments out of tolerance
7.1.15 Appendix 14: Flow for Control of Environment out of spec limits
7.1.16 Appendix 15: Flow for Control of Non-Conformance Lots at Pre-Test Area

CAUTION STATEMENT:
These procedures are defined to the best ways on "how to". Under any circumstances where there is no guide for a specific situation, the Engineer or QA (point of detection of defect) will initiate a meeting to decide action requirement.

7.2 The following sections shall be used for Quarantine/ On-hold, Recapture and MRB disposition of Non-conforming Material/Products.
- 7.2.1 Section A: Procedure for Quarantine/On-hold
- 7.2.2 Section B: Procedure for Recapture and Containment
- 7.2.3 Section C: Procedure for MRB Disposition
- 7.2.4 Section D : Procedure fo QSAT (Quality Sensitivity Alert Tag)

8.0 Safety Precautions:
N/A

9.0 Maintenance/Schedule:
N/A

10.0 Document and Records:

10.1 Following Requires Retention (as minimum):
- i) Recapture data
- ii) Pertinent test or investigation results that assist in the material/products disposition
- iii) MRB form
- iv) Concession form (if applied)
- v) Containment action

NOTE: All the above documents/records shall be retained as per SPEC#: CRQA100G02.

11.0 Others:

11.1 Definition:
- 11.1. Non-conforming material/products are material or products which do not conform to product requirements (customer requirements or specifications). Product with unidentified or suspect status shall also be classified as non-conforming product, e.g. products caused by equipment related events (jams, aborts, mis-process, etc.), mishandled product or material and product without identification.
- 11.1. Corrective action:
  - Action taken to eliminate the cause of an existing non-conformity or other undesirable situation in order to prevent reoccurrence.
- 11.1. MRB
  - Material Review Board

No	Attachment	Title
1	Attachment A-1	On-Hold Card
2	Attachment A-2	Non Conforming Material Tracking Sheet
3	Attachment A-3	Material Quarantine / Off Quarantine Request
4	Attachment B-1	Recapture Sequence
5	Attachment B-2	Recapture Request Form
6	Attachment B-3	Recapture procedure for lot withdrawal from Finish Goods Store
7	Attachment C-1	MRB Issuance Guideline
8	Attachment C-2	Material Review Board
9	Attachment C-3	Reliability Test Risk Assessment Matrix Table
10	Attachment C-4	Flow On Tighten Sampling
10	Attachment C-5	IFN MRB Guideline Checklist Customer Special Requirement (INFINEON)
11	Attachment C-6	IFN LD Tracker Format Customer Special Requirement (INFINEON)
12	Attachment D-1	Quality Sensitivity Alert TAG (QSAT)
13	Attachment D-2	Material Disposition Test Requirement Guideline
14	Attachment D-3	Material Disposition Test Requirement Guideline- Mold Machine Daichi Model (Applicable for MLPN)
15	Attachment D-4	Material Disposition Test Requirement Guideline- Machine Critical Error (Applicable for S site)
16	Attachment D-5	Material Disposition Test Requirement (Applicable for S-Site)
17	Attachment D-6	Material Disposition Test Requirement Guideline – Manual/ Semi-manual Handling for Defect Verification
18	Attachment D-7	Hold Cabinet for hold reason code in MES

Section A

1.0 Title

Procedure for Quarantine / On-hold

2.0 Procedure:

2.1 Any lot placed on-hold at production area shall be clearly identified with "ON- HOLD" tag (refer to form#CRQA100H07-2 - attachment A-1).

2.1.1 Non-conforming lots or suspected non-conforming lots.

2.1.1.1 Upon detection of any non-conforming lots or suspected non-conforming lots, QA personnel (supervisor/Engineer) shall be informed immediately.
Note: If the defect is triggered in production line, Process Engineer shall stop affected operation /package/ material immediately.

2.1.1.2 The lots shall be held on-hold physically at QA cabinet at respective operational area and labeled as "Quarantine" and attached with "ON-HOLD CARD". Lot will be on-hold in QA Quarantine cabinet until received a disposition from customer or completed approved of the MRB.

In the event that QA Quarantine Cabinet is full or there are many lots was on-hold due to one issue:

- Ensure all the on-hold lots are placed in one dedicated place (N2/ CDA cabinet only applicable for unmolded material or subjected to customer specific requirement) and attached with "ON-HOLD CARD" (refer Attachment A-1).
- Fill up the Non-Conforming Material Tracking Sheet (refer Attachment A-2) and attached together with all the on-hold lot physically.

2.1.1.3 Requester needs to specify the hold duration and provide investigation findings before hold duration expires. Extension should be requested if additional time is required. If hold duration expires QA personnel will escalating to requester for lots disposition.

2.1.1.4 The parts placed under quarantine shall be recorded on the quarantine log (refer to form#CRQA100H07-3 - attachment A-2) and QA personnel ensure quarantine log tally between physical lot kept inside cabinet vs quarantine log

2.1.1.5 The lot shall be held electronically at MES system by QA. Auto hold feature will be available for critical defects.

2.1.2 Non-conforming Raw Material

2.1.2.1 If there are raw materials found to be suspected to give problem in production line, the requester shall initiate Material Quarantine / Off Quarantine Request form (refer to form#CRQA100H07-4 - attachment A-3) to IQA to quarantine the material while engineering investigation is being carried out to determine root cause of the problem.

2.1.2.2 Requester need to specify the hold duration and provide investigation findings before hold duration expires. Failure to do so will result in issue escalating to QA Engineer and Product Manager for material disposition.

2.1.2.3 If material is not the cause of problem, then it will be released back to production as per engineer remarks (refer to form#CRQA100H07-4 - attachment A-3). Otherwise, it will be feedback to the supplier and products using the defective parts shall be disposed by MRB (refer to Section C).

2.1.2.4 The affected batches of material shall be held physically at quarantine room or dedicated area and labeled as "Quarantine" or "ON-HOLD".

2.1.2.5 IQA shall also electronically hold the affected batches in IT system
.

2.1.3 Non-conforming Wafers

2.1.3.1 Upon detection of any non-conforming wafer, the lots shall be held on-hold physically at quarantine cabinet by IQC personnel.

2.1.3.2 IQC personnel shall electronically hold the non-conforming wafers in MES system. For some characteristics which are set in the system, auto held feature will activate if beyond the set limit.

2.1.4 Low Yield

2.1.4.1 For lots which do not meet yield trigger limits (low yield) will be physically held at Engineering "ON HOLD" cabinet or rack.

2.1.4.2 These lots will be electronically auto-held in MES system.

2.1.5 Engineering on-hold lots

2.1.5.1 For lots which require engineering attention. (e.g. engineering evaluations, waiting for engineering instructions, engineering follow-up requirements etc.) will be physically held at Engineering "ON HOLD" cabinet or rack.

2.1.5.2 These lots will be electronically held in MES system and physically placed with ON-HOLD card by Engineer.

2.1.6 Problematic lots flow from Assembly to Final Test

2.1.6.1 For the lots which require Final Test attention (eg: waiting for customer instruction, waiting for evaluation result and etc) Process Engineer / Owner will physically placed ON HOLD TAG for those problematic lots to flow from Assembly to Final Test.

2.1.6.2 Test QA will ON HOLD the affected lot in the system once they received the physical lots with ON HOLD TAG.

2.1.6.3 Those lots will be electronically auto held in the system.